0 · Imitating RoboCup locomotion

Original environment → varied surfaces → initialization → representations

Similar walking performance does not imply similar imitation fidelity.

A focused narrative for the final eight-page report. Background and slide notes

1 · Redid experiments — how and why

Why: historical runs used evolving collection, recovery and training procedures, making comparisons harder to interpret.

How: one consolidated protocol for BC, mixed DAgger, GAIL-BC and GAIL-DM:

- Long runs and short turns; shared control interface.
- Explicit allocations and managed synchronous servers.
- Common test rounds, fall handling and metrics.

Outcome: compare training protocols, not different generations of tooling.

2 · Research questions and evidence

Research question	Evidence to prioritize
How stable are the learned controllers?	Falls, joint jerk, posture and useful movement
What do the tested resources buy?	Fidelity and performance at stated data/interaction budgets
Do policies match the expert?	Progress, velocity, tracking and action error together

Surface variation, initialization and context extend these questions. Probing supports their interpretation; it is not a separate transfer claim.

3 · Experimental progression

Stage	What changes?
Original scene	Train and evaluate the four principal methods
Frozen policies	Change slip without updating weights
Varied training	Train across surfaces; hide or supply slip
Remove pretraining	Compare continuation with varied-only learning
Probing	Examine shared-state features and expert similarity
Feedback controls	Match BC optimization; test zero native actions

Training slips: 0.025, 0.1, 0.2, 1, 10. Evaluation: Gaussian draws around these values, fixed throughout each run. Each deployment: 3,000 active steps / 60 simulated seconds.

4 · Standardised allocations

Method	Main allocation
BC	170,000 expert examples
Mixed DAgger	100,000 seed examples + 7 × 10,000 learner-state labels
Each GAIL	170,000-example expert pool + 70,000 PPO interactions

BC: 17 million supervised example exposures. Mixed DAgger: 54 million exposures through repeated aggregate training.

GAIL also inherits the cost of its initializer. Report data, oracle queries, interactions and optimization separately.

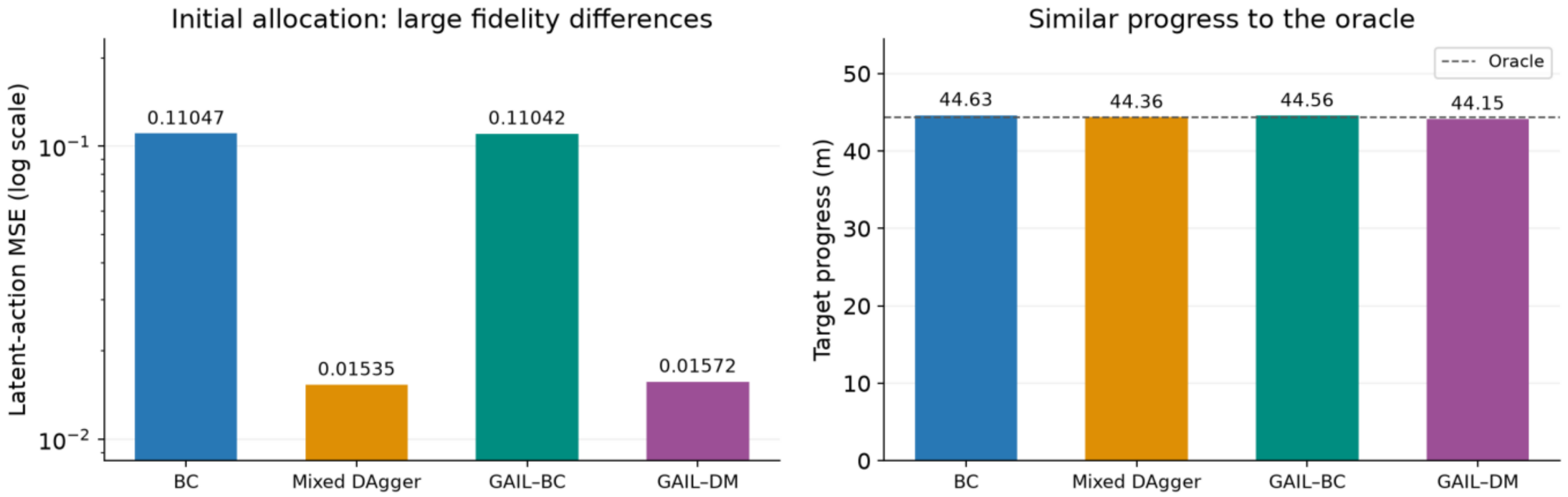
5 · Success measures

Axis	Main measurements
Task/environment performance	Target progress, normalized score, forward velocity, tracking
Imitation fidelity	Held-out latent-action MSE; joint occupancy SWD
Mechanical stability	Falls per active minute, torso RMS, angular jerk
Representation	Held-out probe R^2; expert CKA similarity

Normalized score = (policy progress − random progress) / (expert progress − random progress).

Score 1 means oracle parity on that surface, not unchanged absolute performance.

6 · Original scene: initial allocation

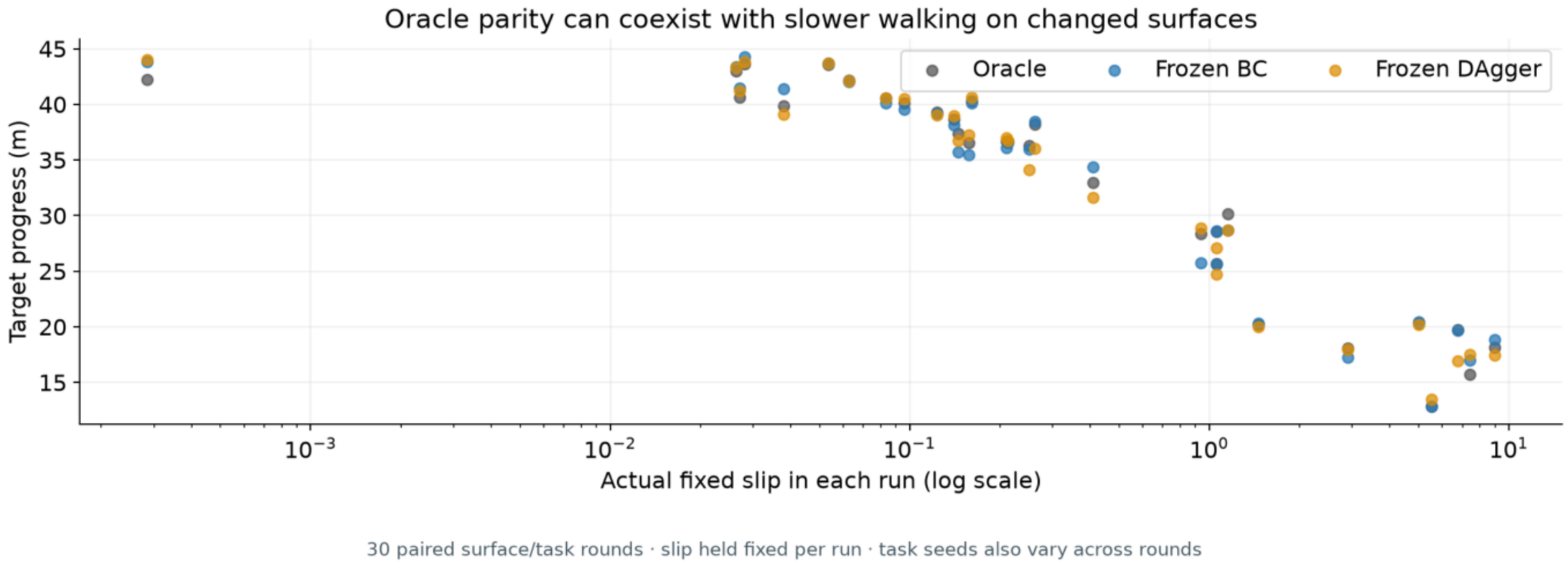


Initial allocations · 5 deployments/controller · BC receives fewer optimizer updates than DAgger

Similar progress conceals large initial fidelity differences.

GAIL endpoints remain close to their initializers. Matching BC updates changes the ranking later (slide 16).

7 · Frozen policies on varied surfaces

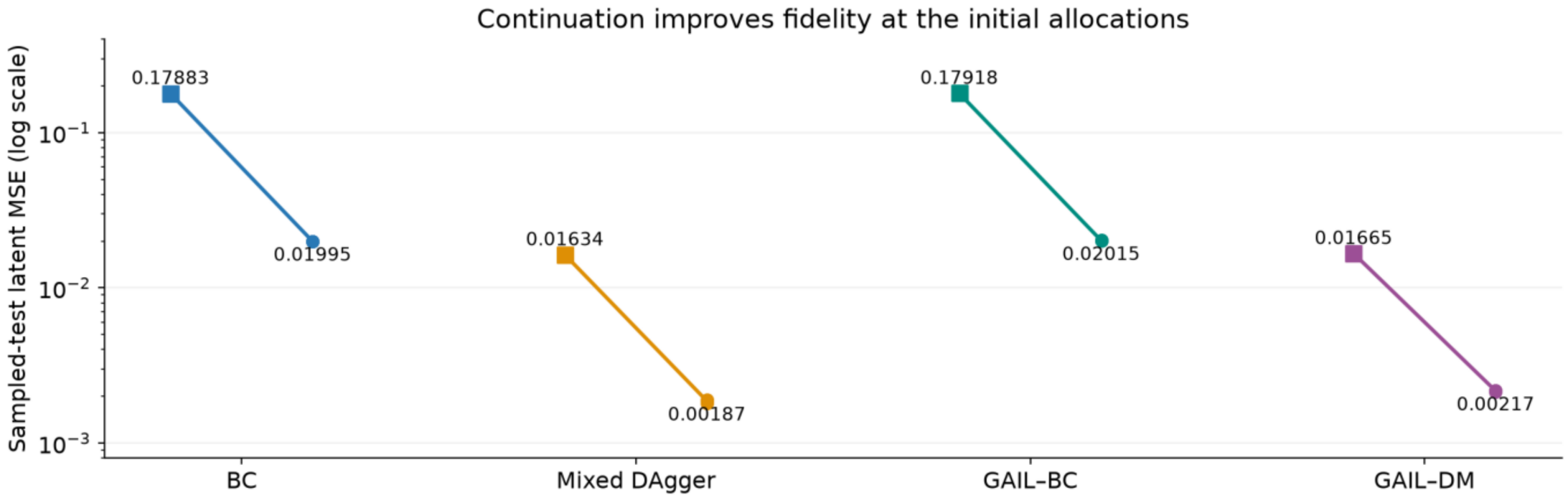


30 paired surface/task rounds · slip held fixed per run · task seeds also vary across rounds

Oracle parity does not mean unchanged walking performance.

Each dot uses its actual fixed slip. Task seeds also vary; this is not a slip-only causal response curve.

8 · Varied training: the initial comparison

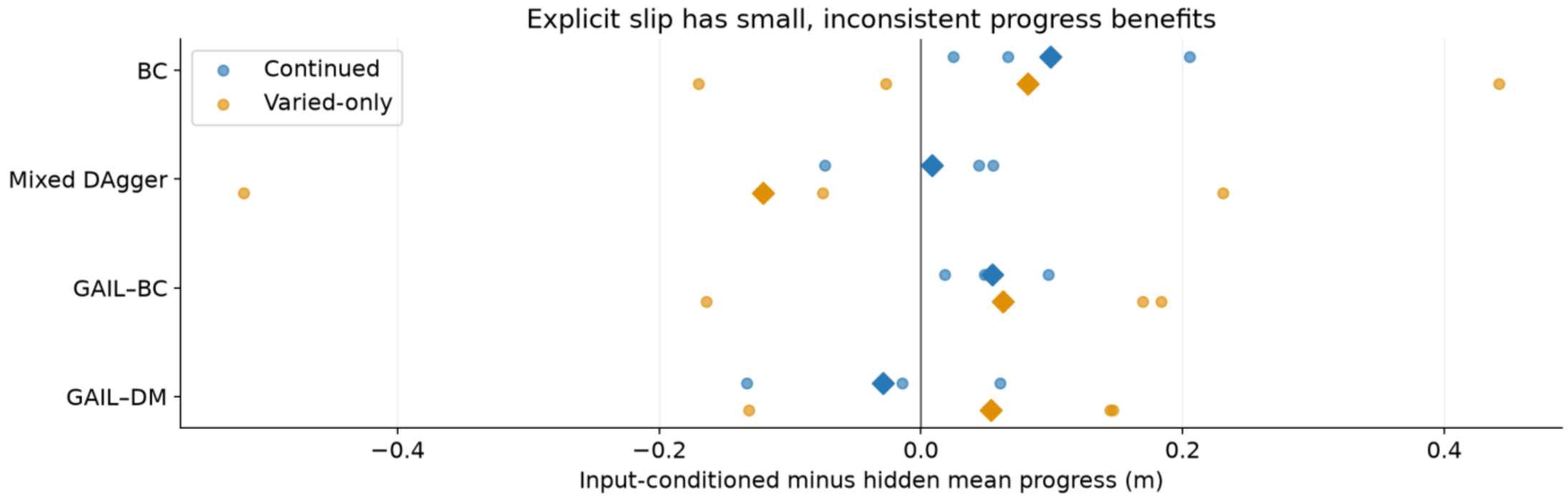


Square: frozen original model · dots: 3 continued hidden models · more training and changed data are confounded

Continuation improves fidelity, but also adds optimization.

Longer original-only BC later improves sampled-surface stability too; surface diversity is not the only explanation.

9 · Explicit slip supply



Dots: per-training-seed differences on shared 30 rounds · diamonds: means · initial allocations, not new 200-epoch fits

Small, inconsistent progress gains at the initial allocations.

The expert can respond to surface effects through ordinary observations without an explicit slip input.

10 · Remove original pretraining

Continuation: original training → varied-domain training.

Varied-only: random BC/DAgger initialization → varied-domain training; GAIL then starts from the corresponding varied-only BC/DAgger.

- Same incremental varied training allocations and evaluation rounds.
- Original policies and completed results preserved and reused.
- Additional direct-random GAIL tests isolate a different starting point.

Important: equal additional budgets do not equalize lifetime training history.

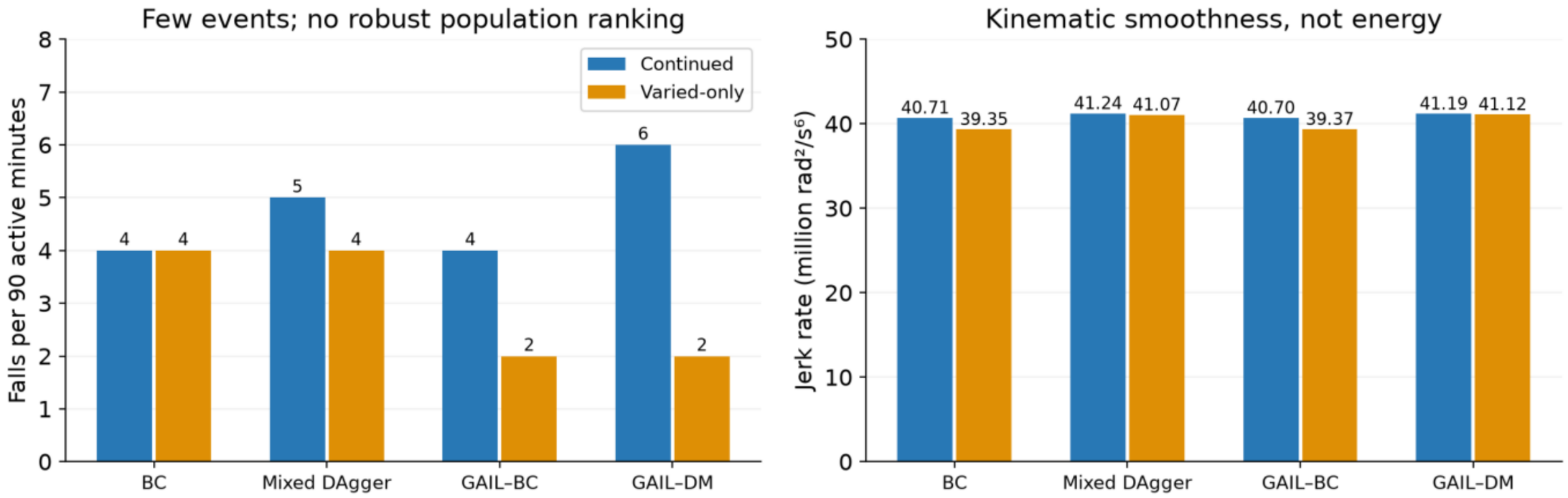
11 · Without pretraining: results and meaning

Hidden-context families; continuation → varied-only scratch:

Method	Progress (m) ↑	Test action MSE ↓
BC	32.784 → 33.192	0.01995 → 0.11177
Mixed DAgger	32.844 → 32.917	0.00187 → 0.01035
GAIL-BC	32.838 → 33.237	0.02015 → 0.11199
GAIL-DM	32.851 → 32.912	0.00217 → 0.01061

Scratch BC gains about 0.41 m, but action error is 5.6× larger. Scratch DAgger still has about 10.8× lower error than scratch BC.

12 · Mechanical results



Hidden models at initial allocations · 3 training replicas × 30 rounds · recovery excluded from active time

Sparse falls do not establish a population ranking.

Jerk measures kinematic smoothness—not torque, energy consumption or hardware wear.

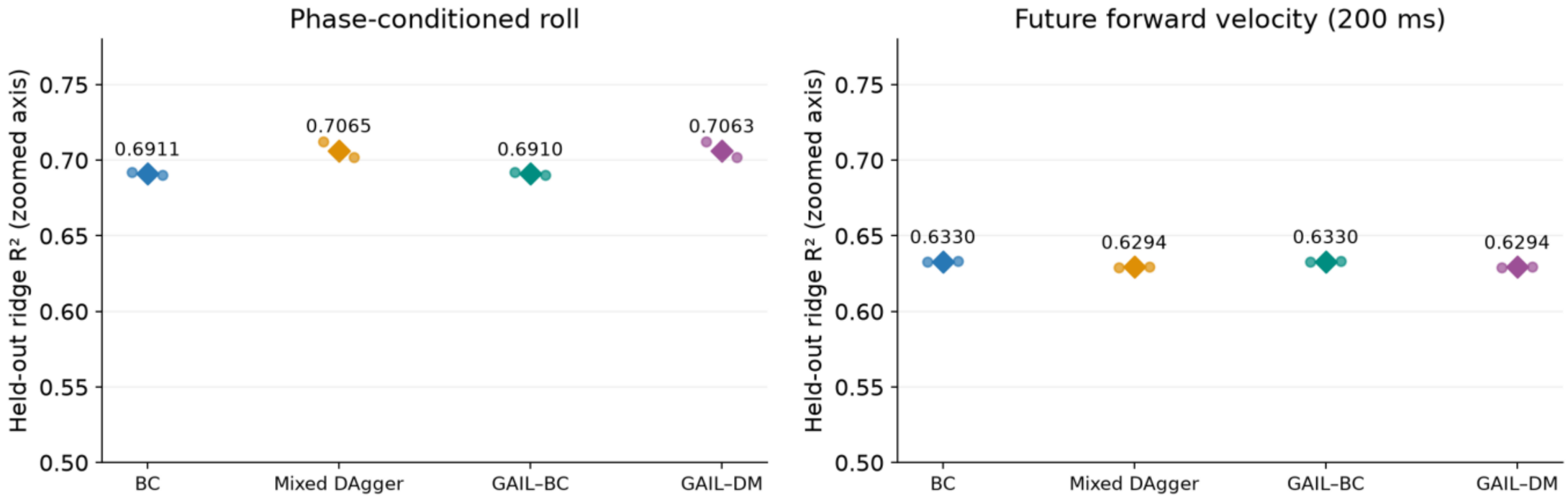
13 · Probes interpret the behavioural results

Earlier finding	What we ask of the representations
Initial-budget DAgger has lower action error	Is phase-conditioned posture more accessible?
Continuation preserves fidelity	What changes when pretraining is removed?
GAIL adds limited improvement	Do its readouts differ from its initializer?

Ridge regression reads physical quantities from frozen features. CKA compares expert/learner feature geometry on shared states.

Held-out episodes and raw/random/shuffled controls distinguish accessibility from mere input information—not causality.

14 · Probing interprets the prior model results



First-layer features on shared oracle states · dots: 3 replicas, diamonds: means · new matched-BC models were not probed

GAIL readouts remain close to their initializers; no single feature explains every outcome.

These are the prior models. The new matched-update BC checkpoints have not been probed.

15 · Pretraining: fidelity is not global similarity

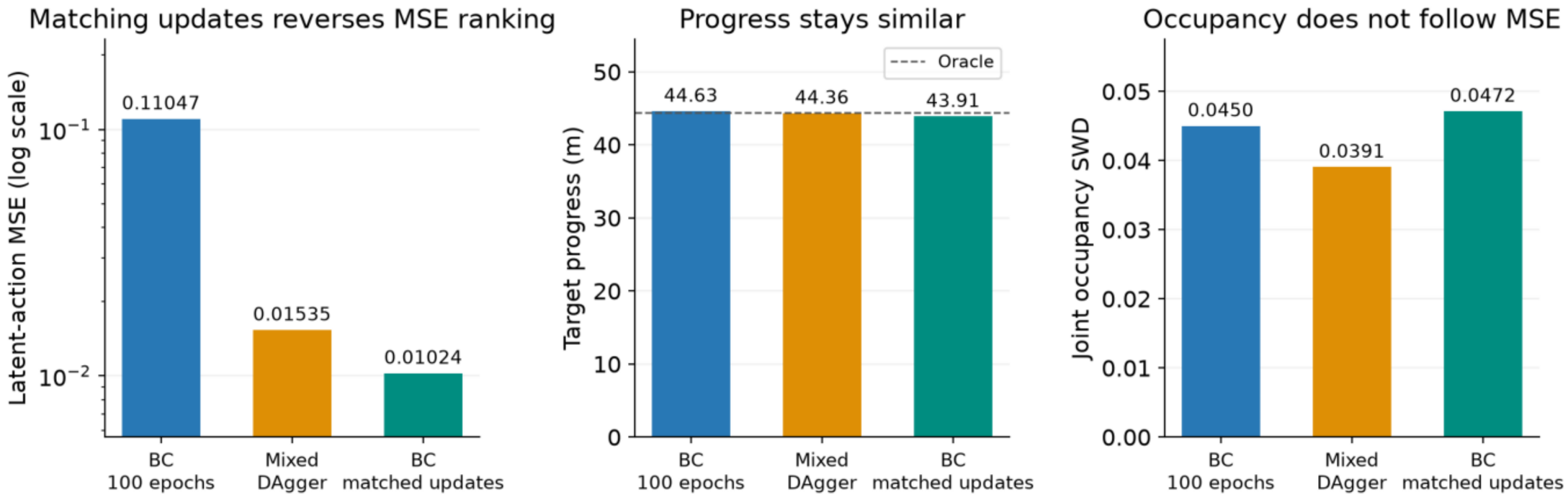
Hidden mixed DAgger; continuation versus varied-only scratch:

Measurement	Continuation	Scratch
Action MSE ↓	0.00187	0.01035
Phase × roll R^2 ↑	0.7065	0.6935
Future-pitch R^2 ↑	0.5876	0.6101
Expert CKA ↑	0.9212	0.9428

Continuation combines closer actions with a better posture readout. Scratch improves other readouts and similarity without improving fidelity.

Interpretation: training history changes what is accessible; “more expert-like features” alone does not explain better imitation.

16 · Feedback: match optimizer updates

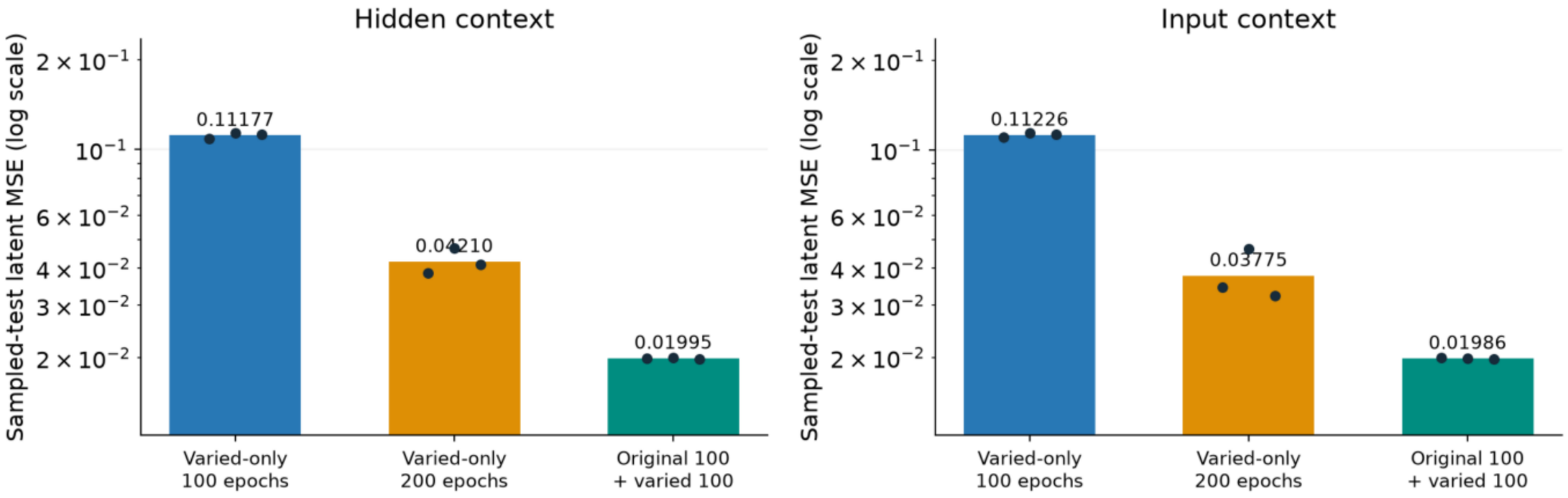


Original scene · 5 deployments/controller · one training seed · means, not superiority tests

BC now has 33% lower MSE than DAgger at 211,050 updates.

The earlier ranking was not an algorithm-only result. Progress and joint occupancy do not improve uniformly.

17 · Feedback: separate exposure from pretraining

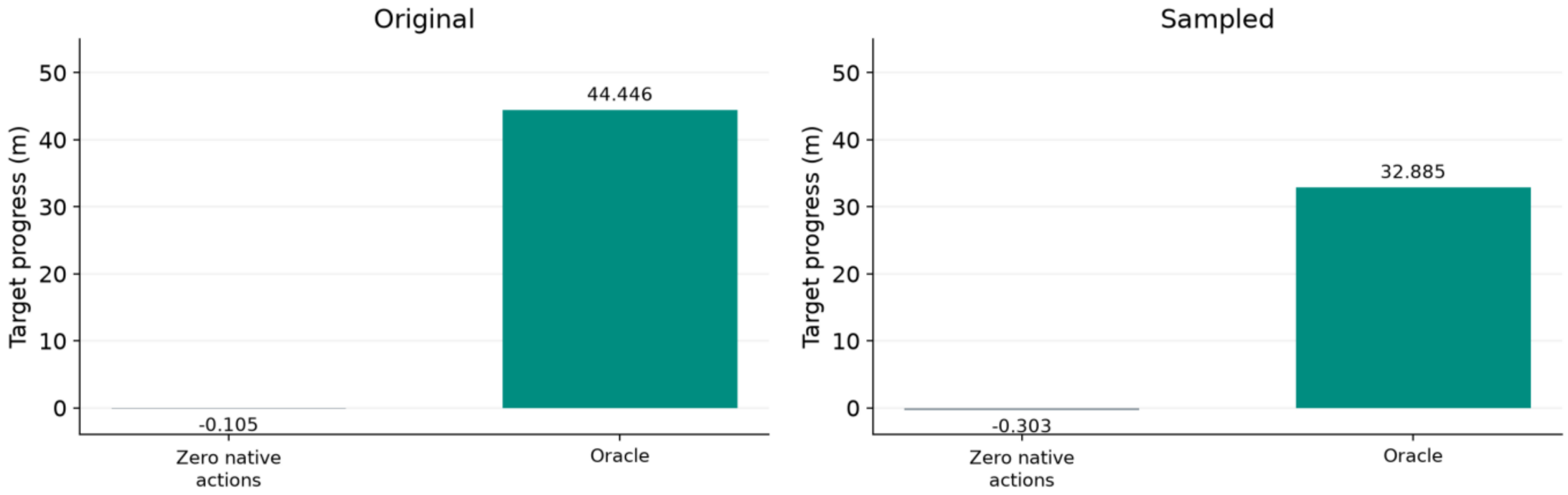


Bars: means; dots: 3 training replicas · last two conditions allocate 34 million exposures; data/history still differ

The gap shrinks from about 5.6× to 2.11× / 1.90×.

Last two conditions match allocated exposures, not data history. Best checkpoints remain near the budget limit: convergence is unproven.

18 · Feedback: does the engine solve the task alone?

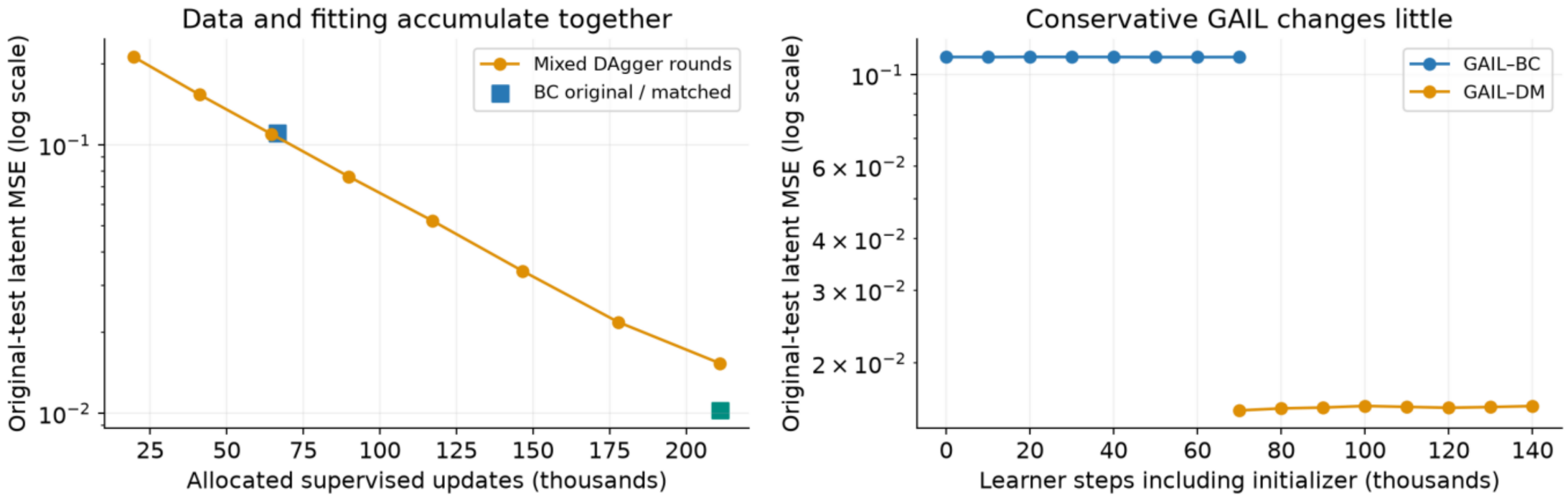


Same gait/IK engine and resets · original: 5 rounds; sampled: 30 · zero actions are not expert-mean actions

Zero native actions do not produce useful progress.

Zero original-scene falls coexist with almost no movement. This does not test every constant action vector.

19 · Feedback: recover fidelity/resource curves



Retrospective fidelity curves, not return convergence · GAIL–DM inherits 70k learner steps; GAIL–BC inherits BC fitting

GAIL refinement changes little under this recipe and budget.

DAgger accumulates both labels and fitting. These are retrospective fidelity curves, not demonstrated return convergence.

20 · Paper conclusions after the controls

- Task parity is not imitation parity: similar progress masks large action-error differences.
- Optimization allocation changes the fidelity ranking: matched BC overtakes DAgger on held-out action MSE.
- Training history matters, but exposure explains part of the gap: longer varied-only BC narrows it substantially.
- Explicit context and conservative GAIL show no consistent broad advantage.
- Probes describe selected old checkpoints; they do not explain the new BC controls or establish causality.

Direct-random GAIL did not learn useful walking at the tested 70k-step budget; this is not a universal limit of GAIL.

21 · Boundaries and proposal deviations

- Matched BC updates/exposures address specific confounds—not total collection cost or convergence.
- Few independent training replicas; continued models share pretrained parents.
- Paired/reused rounds are controls, not additional independent evidence.
- 16 latent gait actions, not the proposed 22 joint velocities; gait/IK remains active.
- Fixed-per-run slip is not spatially or continuously changing terrain.
- Jerk is kinematic, not measured energy; probes are descriptive, not causal.
- Retrospective fidelity/resource curves, not demonstrated return convergence.

Scope: conclusions about these implemented protocols—not universal algorithm rankings.

22 · Outside the paper — worth continuing

Curiosity-driven direction	Question left open
Wider-slip stress tests → context-aware RL	Where does the oracle fail, and can reward-driven learning go beyond imitation?
Larger-budget GAIL: random, BC, mixed DAgger starts	Can less constrained refinement improve on its initializer?
R1 transfer and representation interventions	Which source-policy properties actually help another robot learn?

These are separate follow-on studies, not evidence for the current paper.

Stress and GAIL pipelines are implemented; RL extensions remain conditional. R1 work is separate and has no established transfer result here.